Biomedical National Elemental Imaging Resource Co-Registration Tool Facilitates Metals-Based Pathway Analysis

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ABSTRACT

- Trace elements play pivotal roles across many biological processes
- Disruptions metal homeostasis → novel biomarkers/therapeutics
- Spatially resolved metal analysis– localize metal bioaccumulation
- Developed BNEIR co-registration tool to facilitate annotation of tissue regions in elemental imaging data from H&E stained WSI
- Applied BNEIR co-registration tool to co-register spatial elemental abundance with spatial transcriptomics co-registered with H&E
- Identifies spatial cell-type, metal, histological and transcriptomic signatures involved with **tumor immune microenvironment** and **nodal/distant metastatic cancer**, further work needed

INTRODUCTION

- **Tumor metastases:** responsible more than 90% of cancer deaths
  - Colorectal cancer (CRC) and breast cancer have a combined annual incidence of more than 400,000 cases and the 5-year survival rate for CRC and breast cancer is 64% and 90%
  - Improvements in screening, prognostication, disease management, and therapeutic options, which are accurate, faster, low-cost are needed
- **Metals:** play an essential role in regulating metabolic programming
  - In breast and colon cancer, copper is heavily implicated in uncontrolled cell proliferation, dynamic cellular signaling and coordinates a host of biological activities
  - Inflammatory response can promote copper metabolism in tumor cells, enabling further tumorigenesis
  - e.g., Cadmium → migration of colon cancer cells
- **Spatially resolved metal analysis (elemental imaging, EI)**
  - Provides detailed, quantifiable multi-elemental distribution maps.
  - Bulk analysis often discards important tissue contexts
  - Challenges remain studying EI by tissue architecture within H&E
- **Goals:**
  - Co-register H&E images with elemental maps to facilitate transfer of annotations and other spatial molecular modalities
  - Integrate/identify correspondence between spatial transcriptomic patterns with elemental distributions around tumors

METHODS

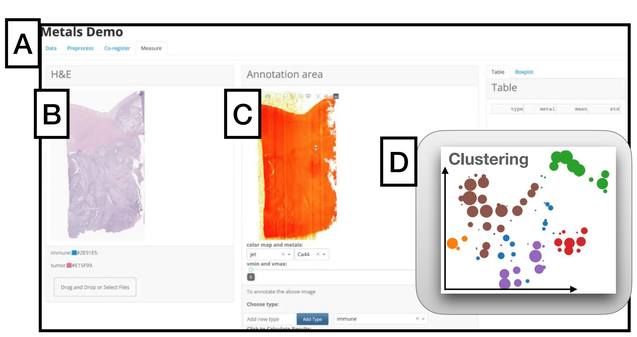

Figure 1: A) Prototype Co-Registration Tool, **B)** H&E WSI from ST experiment, **C)** Elemental maps, **D)** Co-Registration allows for spatial transcriptomics integration

METHODS

- **LA-ICPTOF-MS:** Laser ablation inductively coupled plasma time of flight mass spectroscopy is a spatially-resolved metal analysis technique (also known as elemental imaging)
  - Images all of the elements at 1 micron resolution
  - Biomedical National Elemental Imaging Resource has one of fifteen such devices worldwide, first in United States
- **Spatial Transcriptomics (ST):**
  - Traditional transcriptome-wide association studies lack spatial context.
  - 10X Genomics Visium Spatial Transcriptomics assay allows for high multiplexing at high spatial resolution.
  - Adoption of innovative protocols, CtytAssist– streamline/standardize sample preparation, allowing for operation on archival FFPE blocks.
  - Expression profiles for spots within sections deconvolved into their constituent cell type proportions using single cell RNASeq data
- **BNEIR Co-Registration Tool:**
  - Spatial transcriptomics and LA-ICPTOF-MS sections co-registered via ST CytAssist H&E section, using BNEIR Co-Registration Tool
  - ST data aligned to same slide H&E via CytAssist device
  - A series of corresponding macroarchitectural landmarks placed on the serial H&E slides to measure registration accuracy
- **Gene-Metal-Cell-Type Associations:**
  - Genes and elements filtered by detection, abundance and variance
  - **Hotspot Analysis:** Wilcoxon non-parametric tests for differential expression based on regions defined by Local Getis Ord Statistics, defining regions high in elemental abundance
  - **MEFISTO:** Spatially-aware factor model that derives profiles of genes, cells, histological structures and metals
- **Identify metal modified metastasis-related pathways:** Enrichr

RESULTS

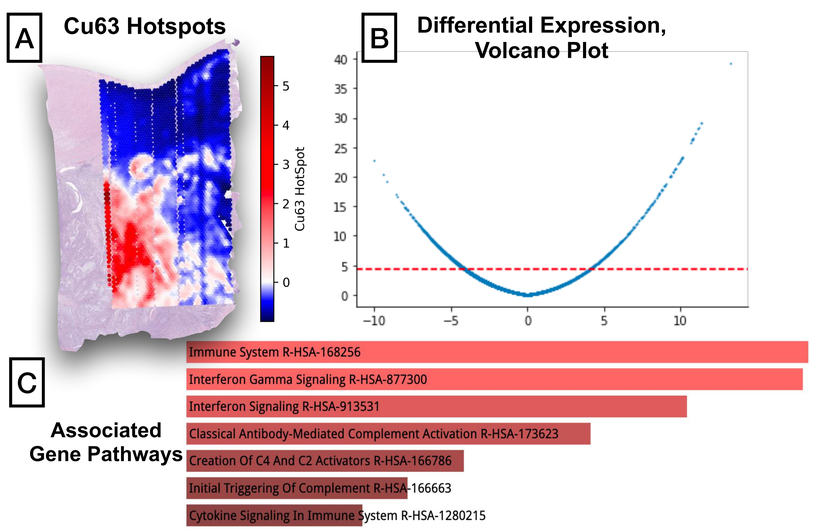

Figure 2: A) Hotspots of Cu63 determined using Getis Ord Statistics, **B)** Differential expression analysis across ~18,000 genes for regions with high/low Cu63 abundance, reveals **C)** immune activation pathways within tumor associated with Cu63

RESULTS

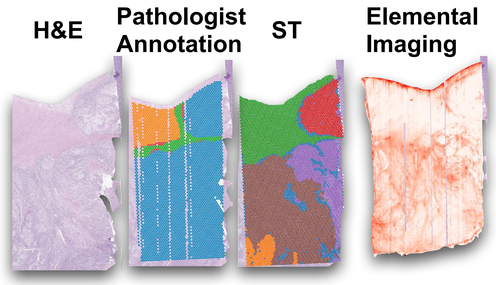

Figure 3: Co-Registration facilitates assessment integrating: **A)** H&E slides to define **B)** Tissue architectures, along with **C)** ST and **D)** EI data

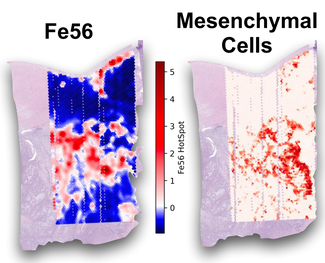

Figure 4: Associations between Iron via hotspot analysis and mesenchymal cells via scRNASeq

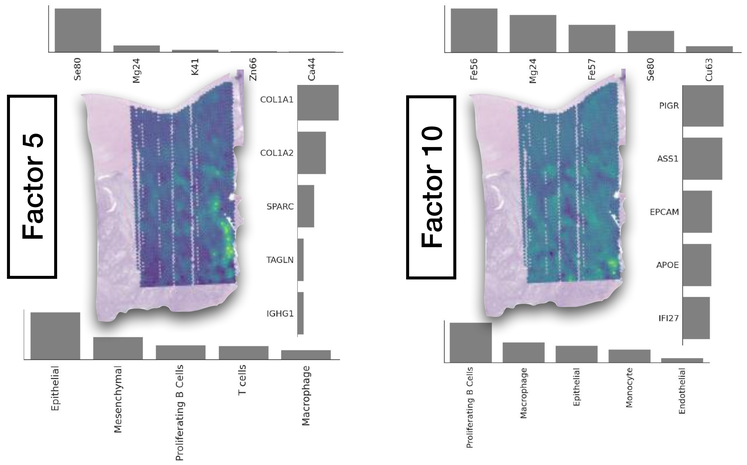

Figure 5: A) MEFISTO Factor 5, linked to EMT **B)** MEFISTO Factor 10, linked to cell proliferation

CONCLUSION

- **Key Research Findings:**
  - **Co-Registration tool** enables joint analysis tissue histology, metals, localizes bioaccumulation to specific architectures
  - **Facilitates integration with ST data**– initial integration revealed associations between immune activation within tumors and copper and mesenchymal signatures with iron
  - **Spatial factor analysis** on multimodal spatial data reveals pro-tumorigenic co-occurring cellular, genes, metal signatures
- **Future Direction:** Diverse tissue contexts prognostic signatures
  - Potential clinical applications in copper depletion therapeutics
- **National Biomedical Elemental Imaging Resource:** BNEIR is a facility for elemental imaging of biological specimens, open to biomedical researchers with NIGMS-supported work. The aim is to support basic research for disease diagnosis, treatment, and prevention. **Efforts to develop software to facilitate integration of IHC, histology, elements, ST for multimodal assessment.**
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Biomedical National Elemental Imaging Resource BNEIR edit.